

Soumen Laha

Aspiring Software Developer | B.Tech ECE, IEM Kolkata

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Summary

Results-driven 2nd-year B.Tech (ECE) student at IEM Kolkata with a strong foundation in **Data Structures & Algorithms**, **Object-Oriented Design**, and **Java/Python**. Experienced in building responsive web applications, embedded IoT systems, and ML-driven solutions. Seeking a **Software Engineer / SDE** role at a product-based company (Google, Amazon, Microsoft) to solve large-scale engineering challenges through clean, efficient code and scalable system design.

Education

Institute of Engineering & Management (IEM) – MAKAUT <i>B.Tech, Electronics & Communication Engineering (ECE) CGPA: 8.5</i>	Kolkata, West Bengal <i>July 2024 – July 2028</i>
Simplalpal MM High School – WBCHSE <i>Higher Secondary, Science (PCM + Biology) 90%</i>	Bankura, West Bengal <i>2022 – 2024</i>
Fulkushma High School – WBBSE <i>Secondary Education 78%</i>	Bankura, West Bengal <i>2022</i>

Technical Skills

Languages: Java, Python, C **OOP / OOD:** Encapsulation, Inheritance, Polymorphism, Abstraction, SOLID Principles
DSA: Arrays, Linked Lists, Stack, Queue, Heap / Priority Queue, Trees, BST, Trie, Graphs, Hashing
Algorithms: Binary Search, Bit Manipulation, Dynamic Programming, BFS/DFS, Greedy, Backtracking, Two Pointers, Sliding Window, Sorting, Recursion, Time & Space Complexity Analysis
Web & Dev Tools: HTML5, CSS3, JavaScript, Responsive Design, REST API, Git, GitHub, VS Code, Linux/CLI
AI / ML: Machine Learning, Supervised Learning, NumPy, Pandas, IoT, Generative AI (ChatGPT API)
Engineering Tools: SolidWorks (3D CAD), MATLAB, Embedded C, Unit Testing, Debugging, Agile/SDLC

Experience

IEM IEDC – Innovation & Entrepreneurship Development Cell <i>Project Intern – Robotics & Mechanical Domain</i>	Kolkata, Salt Lake <i>July 2025 – Present</i>
- Designed and fabricated functional prototypes using SolidWorks CAD and 3D printing; reduced design-to-prototype cycle time by iterative testing and version control of design files. - Collaborated in an Agile cross-functional team to deliver hardware solutions for real-world engineering problems within sprint-based timelines.	
LeetCode – Competitive Programming <i>Algorithm Problem Solver</i>	Remote <i>Jan 2025 – Present</i>
- Solved 100+ problems across Arrays, Linked Lists, Stacks, Queues, Heaps, Trees, Graphs, DP, and Backtracking in Java with consistent focus on optimal time/space complexity. - Mastered interview patterns: Binary Search, Bit Manipulation, Two Pointers, Sliding Window, BFS/DFS, and Greedy for FAANG-level problem solving.	

Projects

Personal Portfolio Website <i>HTML5, CSS3, JavaScript, REST API</i>  View on LinkedIn	2025
- Built a fully responsive portfolio using semantic HTML5, Flexbox/Grid, and vanilla JavaScript; implements mobile-first design with cross-browser compatibility and fast load optimization. - Applied OOD principles for modular JS components; integrated external REST APIs for dynamic content rendering.	
Train Accident Prevention System <i>Embedded C, IoT, Sensor Integration</i>	2024
- Engineered an obstacle-detection system using IR/ultrasonic sensors with interrupt-driven real-time signal processing and automated safety alerts – reducing potential accident response time. - Wrote and unit-tested embedded C modules for sensor calibration and edge-case handling under fault conditions.	
Atmospheric Water Condensation Model <i>Hardware Prototype, Sustainable Engineering</i>	2024
Designed a Peltier-based atmospheric water harvesting system; documented technical specifications and presented as a sustainability engineering solution at IEM's innovation showcase.	

Certifications

Google AI Essentials – Google	2024	Machine Learning with Python – IBM / Coursera	2024
Introduction to Artificial Intelligence (AI) – IBM / Coursera	2024	Introduction to IoT – Coursera	2024
Accelerate Your Learning with ChatGPT – Coursera			2024

Achievements

- Achieved **90%** in Class 12 Science (PCM+Biology) – top academic percentile, Bankura district, West Bengal.
- Completed **5 industry certifications** (Google, IBM) in AI, ML, and IoT alongside full-time B.Tech coursework.
- Active member of **IEM IEDC Robotics Lab**; contributed to prototyping and innovation projects in the first year itself.